ANNA UNIVERSITY

UNIVERSITY COLLEGE OF ENGINEERING- DINDIGUL

DINDIGUL - 624622

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

EE3028 – DESIGN OF ELECTRIC VEHICLE CHARGING SYSTEM

Name	**:**
Register No.	**:**
Branch	**:**



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BONAFIDE CERTIFICATE

This is to certify that this is a bonafide record of work done by Mr./Ms. _____in the _____semester of _____ year in _____ in the Department of Electrical and Electronics Engineering under AU Regulation 2021 during the academic year 2024– 2025.

University Registration No | |

STAFF IN CHARGE **HEAD OF THE DEPARTMENT**

Submitted for the University Practical Examination held on _____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

EE3028 DESIGN OF ELECTRIC VEHICLE CHARGING SYSTEM

L	T	P	C
2	0	2	3

COURSE OBJECTIVES:

- To know the charging station and standards
- To learn the concepts of power converters in charging
- To find the charging scheme in renewable based EV charging
- To demonstrate the wireless power transfer technique
- To design & simulate power factor correction circuits

LIST OF EXPERIMENTS:

1. Simulation and analysis for bi-directional charging V2G and G2V.
2. Design and demonstrate solar PV based EV charging station.
3. Simulate and infer wireless power charging station for EV charging.
4. Simulation of boost converter based power factor correction.

TOTAL: 30 PERIODS

COURSE OUTCOMES:

Upon the successful completion of the course, students will be able to:

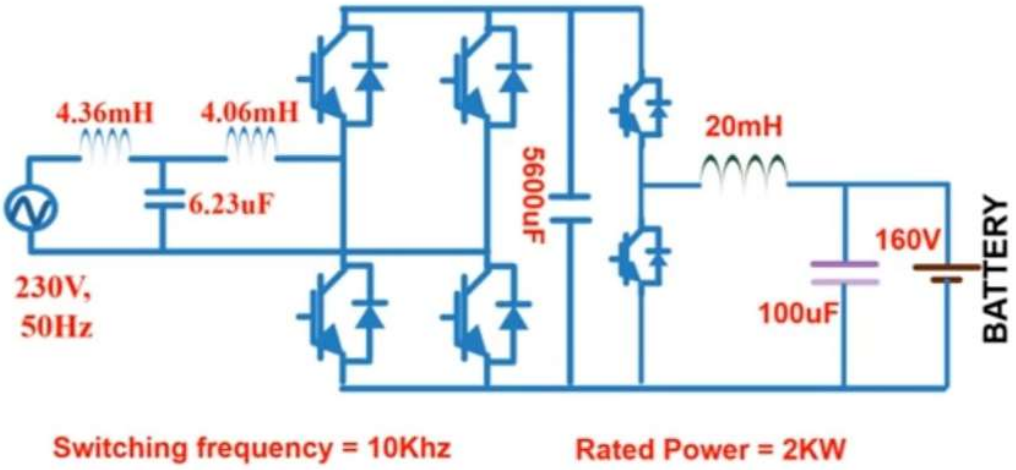
CO1: To illustrate various charging techniques and to know charging standards and regulations.
CO2: To demonstrate the working of DC-DC converters used for charging systems and principles
CO3: To illustrate the advantages of renewable system based charging systems
CO4: To demonstrate the principles of wireless power transfer.
CO5: To analyze the standards for wireless charging
CO6: To design and simulate boost converter based power factor correction.

INDEX

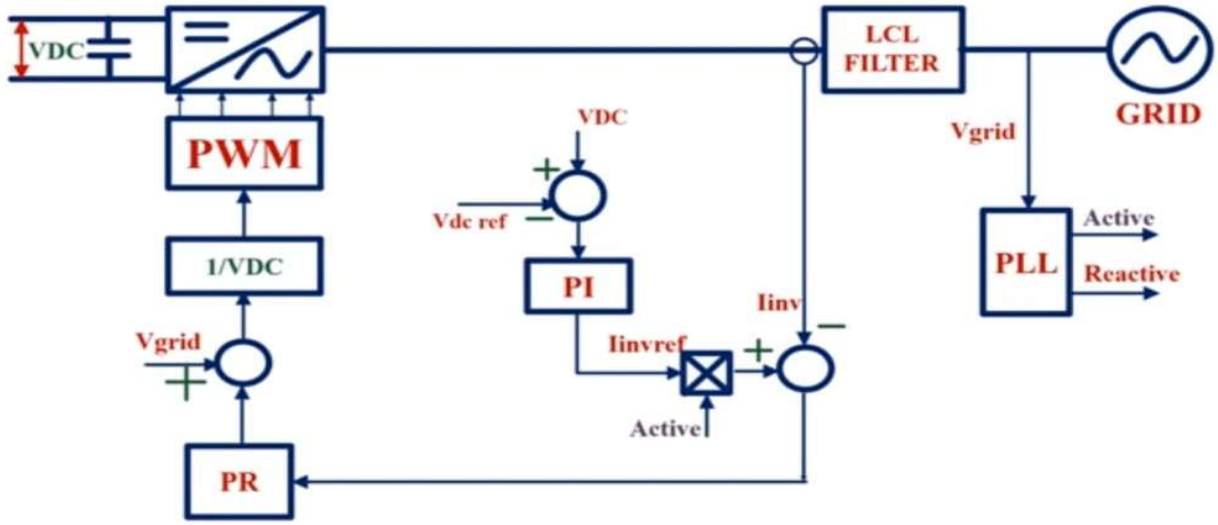
Expt. No.	Date	Name of the Experiment	Page No.	Signature

SIMULATION EXPERIMENTS

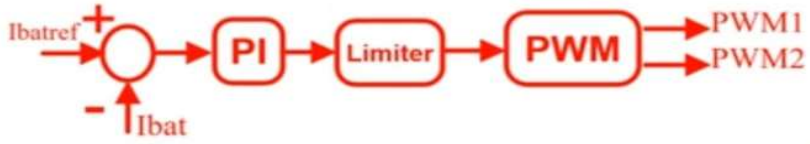
CIRCUIT DIAGRAM:

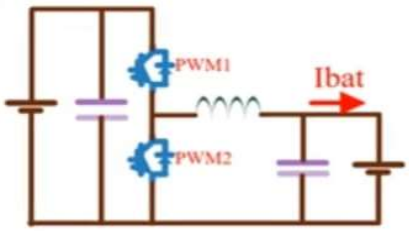


FRONT END CONVERTER CONTROLLER:

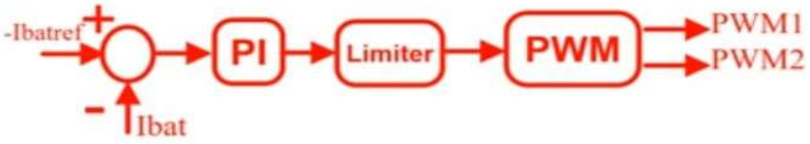


BATTERY CHARGING





BATTERY DIS-CHARGING



EX. NO. DATE:	SIMULATION AND ANALYSIS FOR BI - DIRECTIONAL CHARGING V2G AND G2V	

AIM:

To simulate and analyze bidirectional charging operations, focusing on Vehicle-to-Grid (V2G) and Grid-to-Vehicle (G2V) modes using MATLAB/Simulink.

SOFTWARE REQUIRED:

MATLAB software with Simulink toolbox

THEORY:

Bidirectional charging allows electric vehicles (EVs) to function as both energy consumers and providers in relation to the power grid. This involves two main modes:

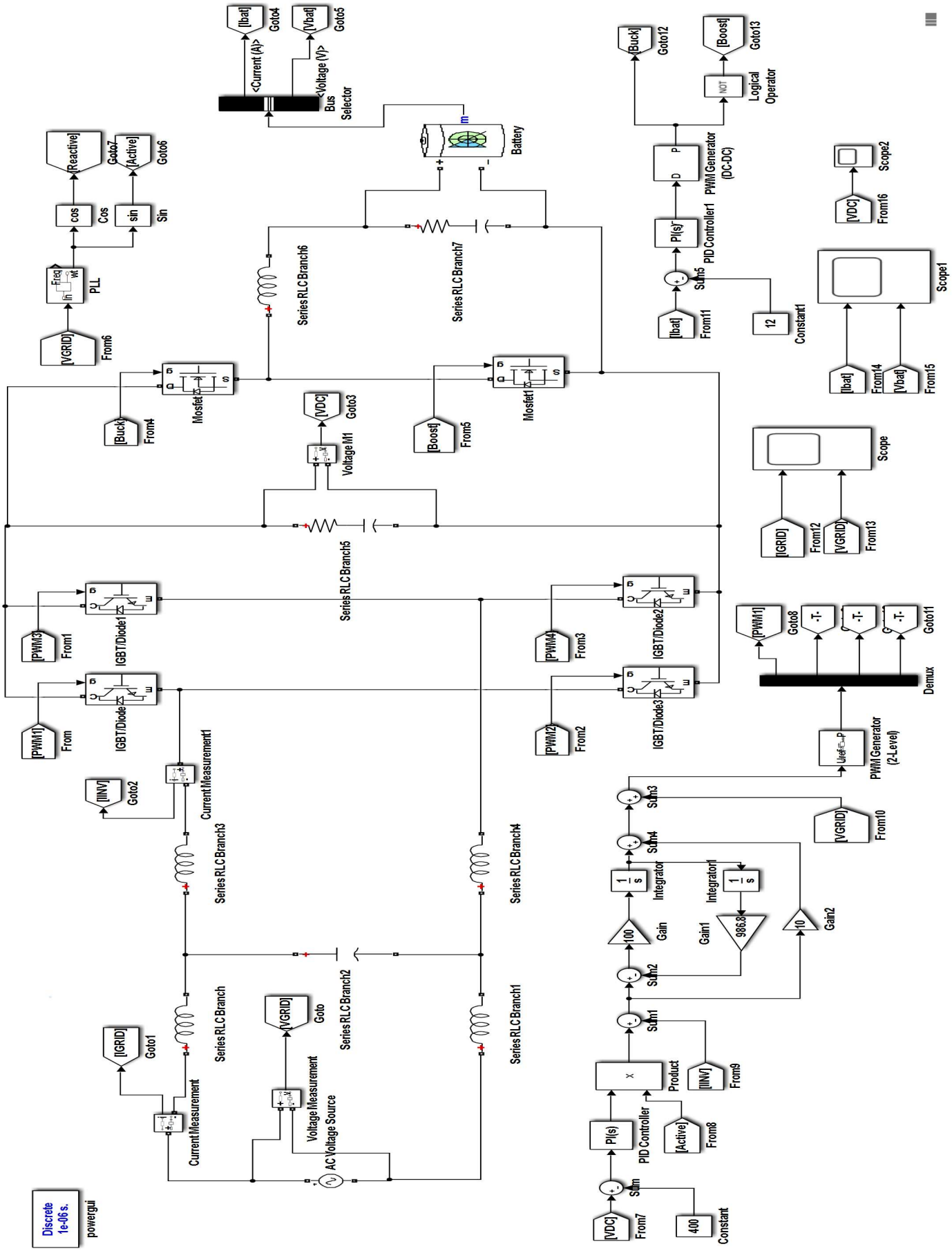
I. Grid-to-Vehicle (G2V) Mode:

- In the G2V mode, the EV is charged from the power grid.
- This mode is designed to allow the battery to absorb energy from the grid during off-peak hours when electricity demand is lower, and supply is ample.
- During this charging process, energy flows from the grid to the vehicle battery, filling the battery's state of charge (SOC).
- Efficient G2V charging requires careful control of current and voltage to maximize battery life and ensure safe, optimal charging rates.
- Additionally, G2V mode can be programmed to draw power when grid electricity costs are lower, reducing operational costs for EV owners and balancing demand across the grid.

II. Vehicle-to-Grid (V2G) Mode:

- In the V2G mode, the EV battery serves as a source of power for the grid, allowing energy to flow back from the EV to the power system.
- V2G is particularly valuable during peak load periods when grid demand is high, providing additional energy to support grid stability and reduce stress on power plants.
- By supplying stored energy back to the grid, V2G can assist in load leveling, frequency regulation, and grid stability.
- This bidirectional flow is achieved through advanced power electronic converters and control algorithms, which ensure seamless transitions between charging and discharging.
- V2G also offers financial benefits, as EV owners can receive compensation for the energy returned to the grid, enhancing the value of their electric vehicle as a distributed energy resource.

SIMULATION DIAGRAM:



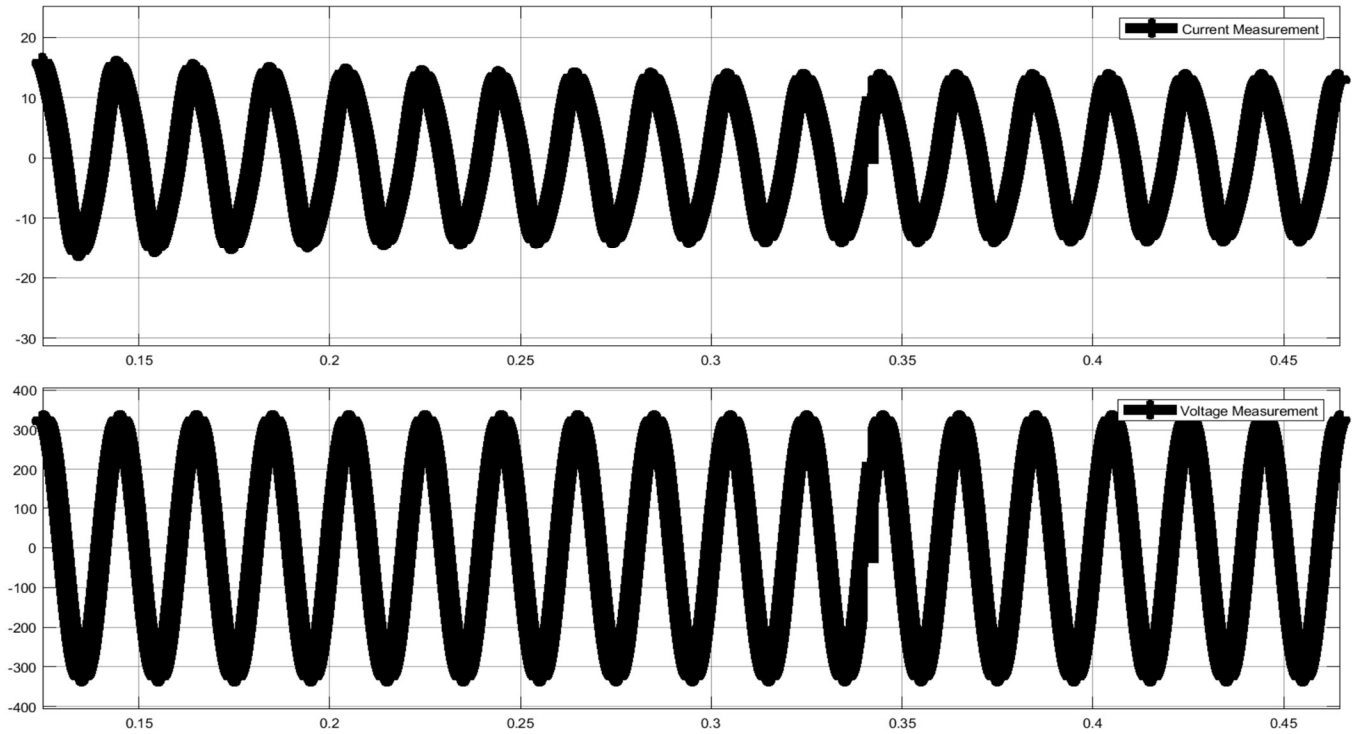
SIMULATION PROCEDURE:

1. Open MATLAB and create a new Simulink model.
2. Add blocks for the power grid, EV battery and bidirectional converter.
3. Create a control block (PI or PID) to manage power flow.
4. Set Initial conditions for all the blocks
5. Run the simulation
6. Now, observe power flow, SOC change and grid interactions during G2V and V2G cycles

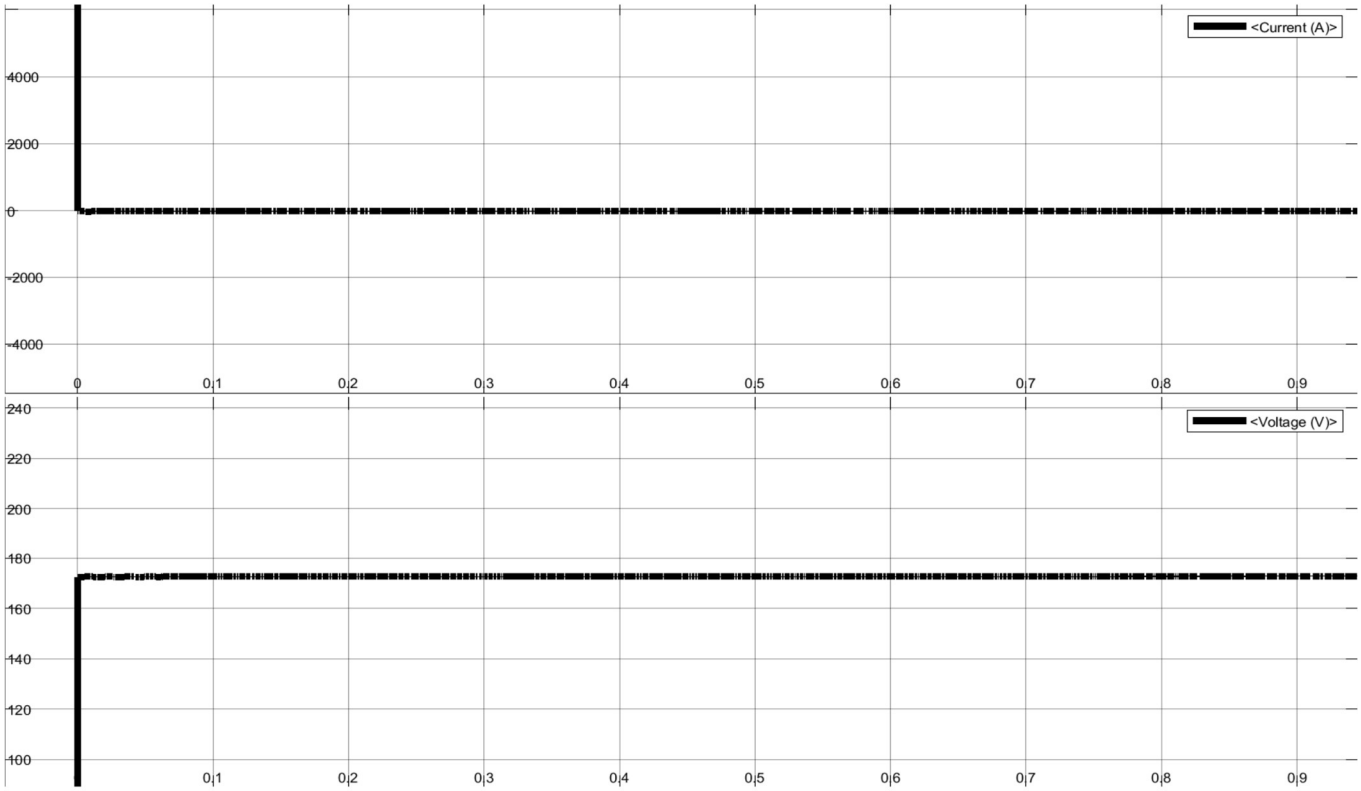
WAVEFORMS:

Case 1: Constant = -12

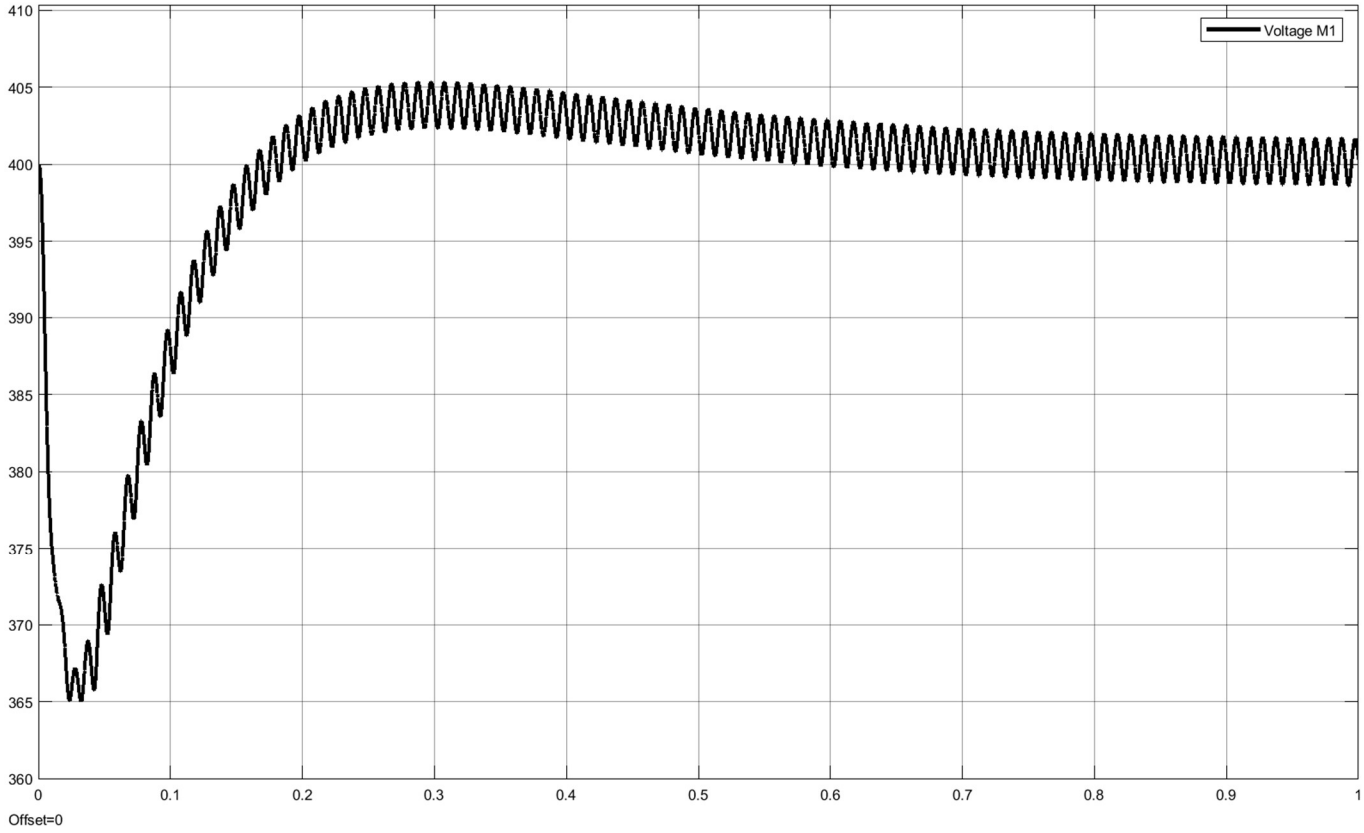
Grid



Battery

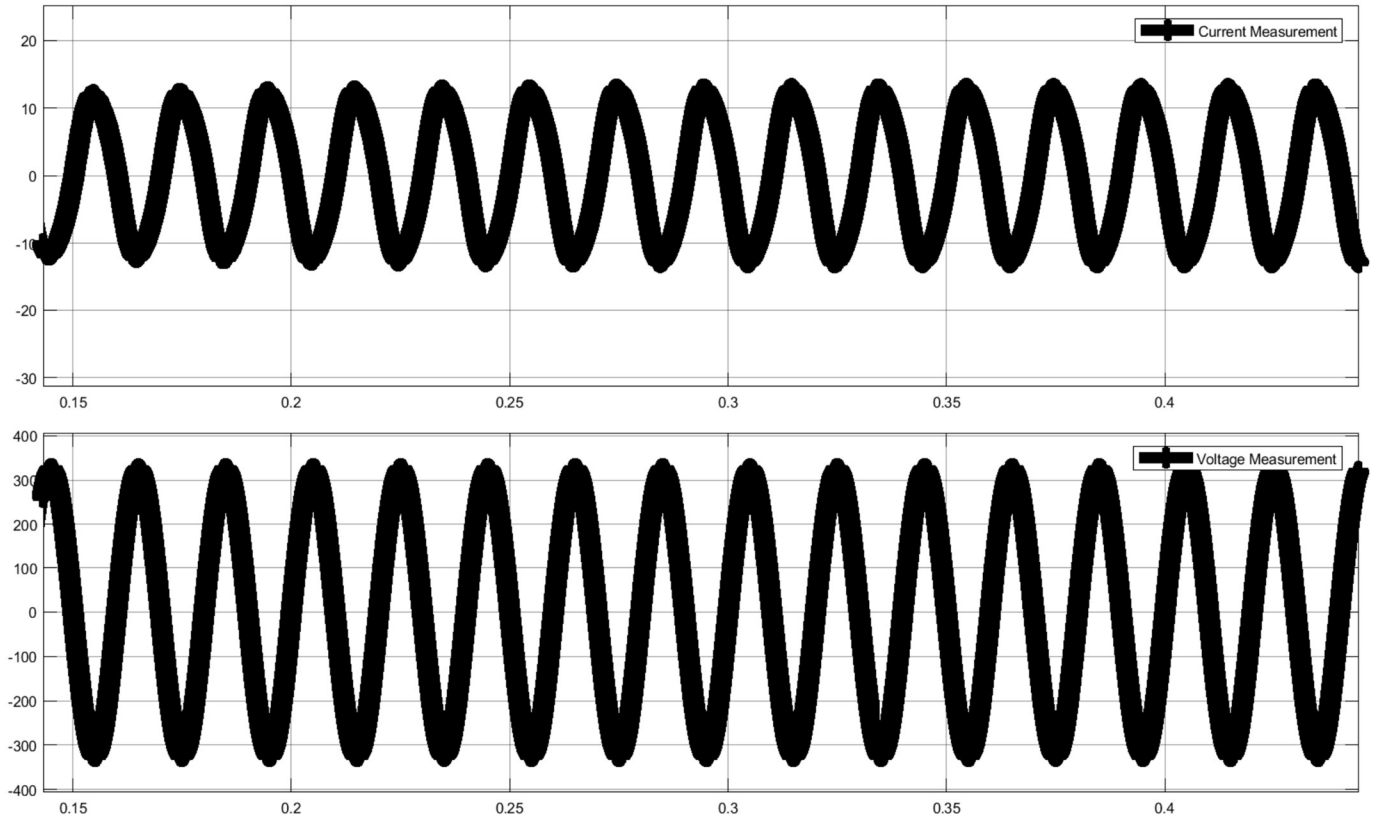


SOC

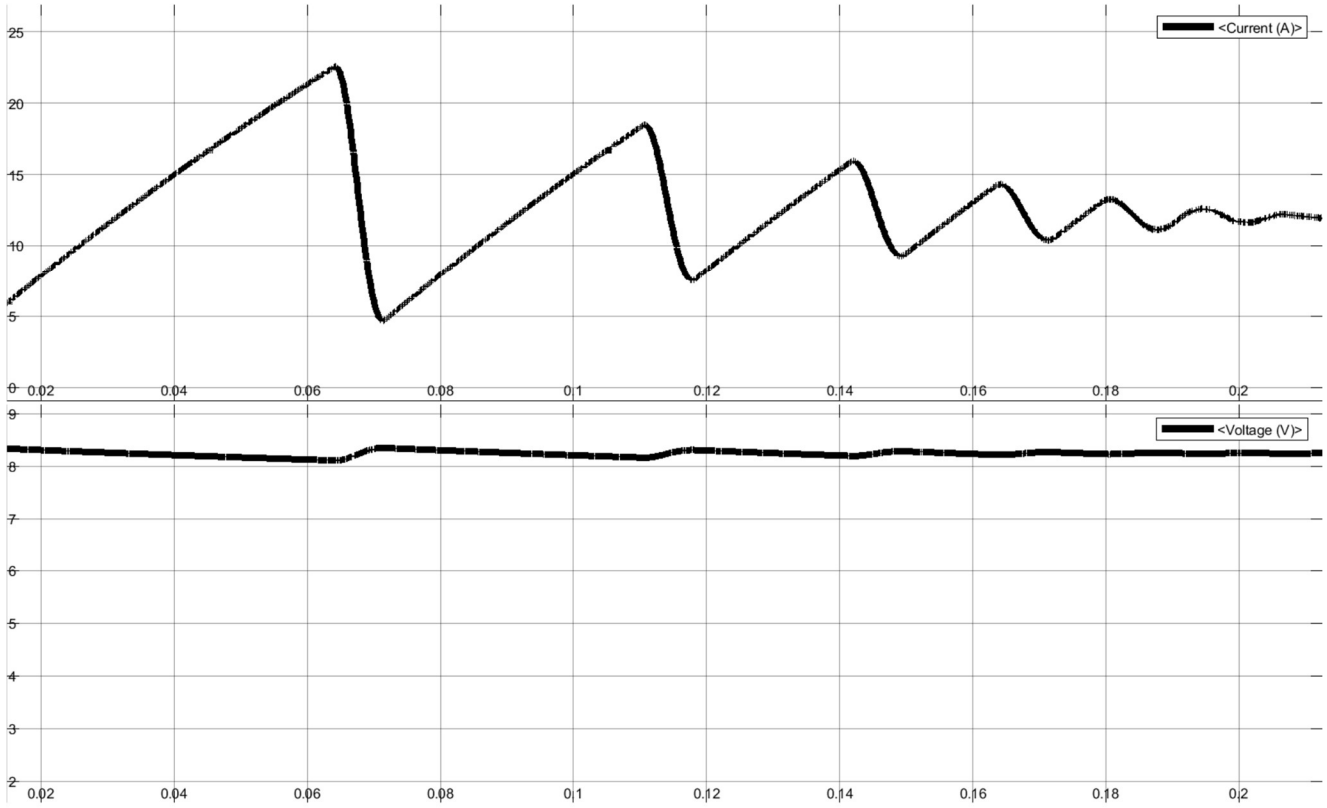


Case 2: Constant = 12

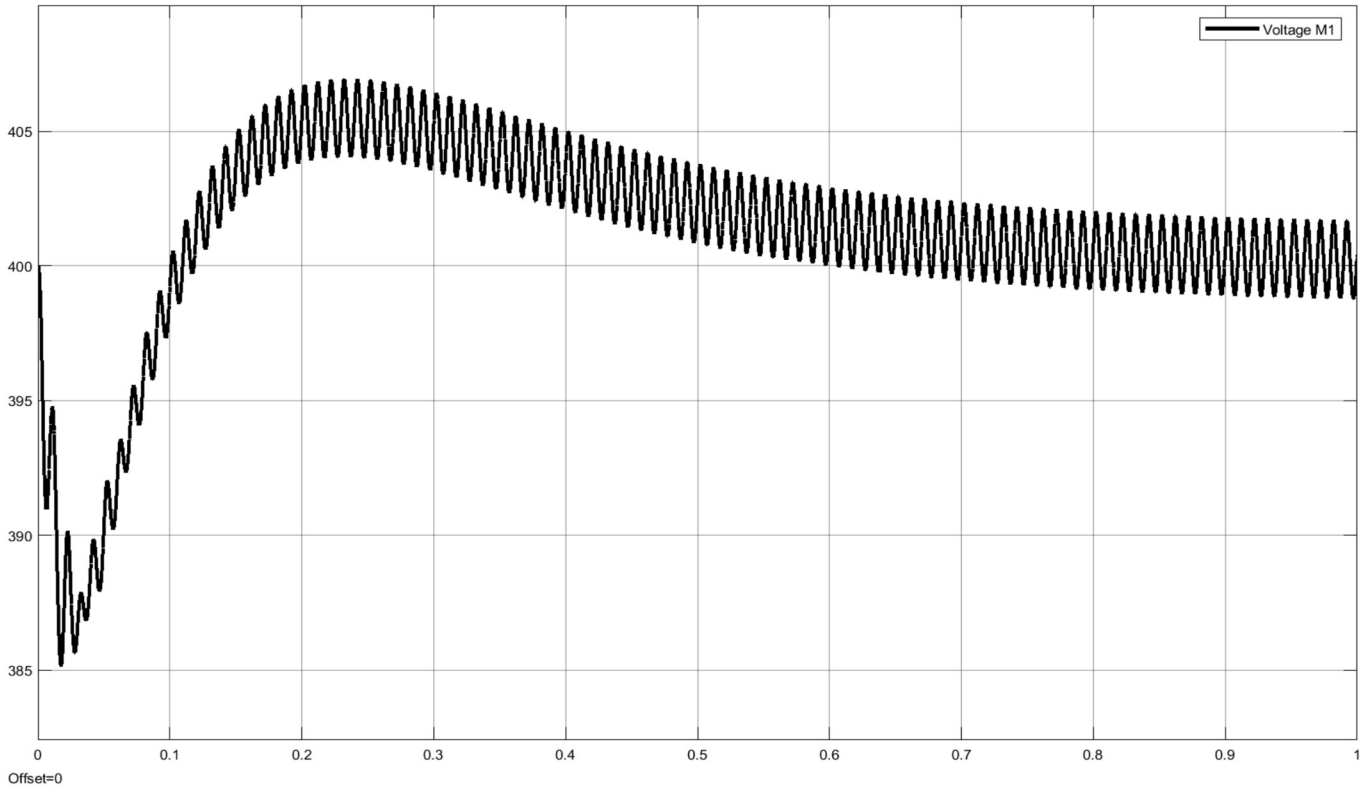
Grid



Battery


<Current (A)>
25
20
15
10
5
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
<Voltage (V)>
9
8
7
6
5
4
3
2

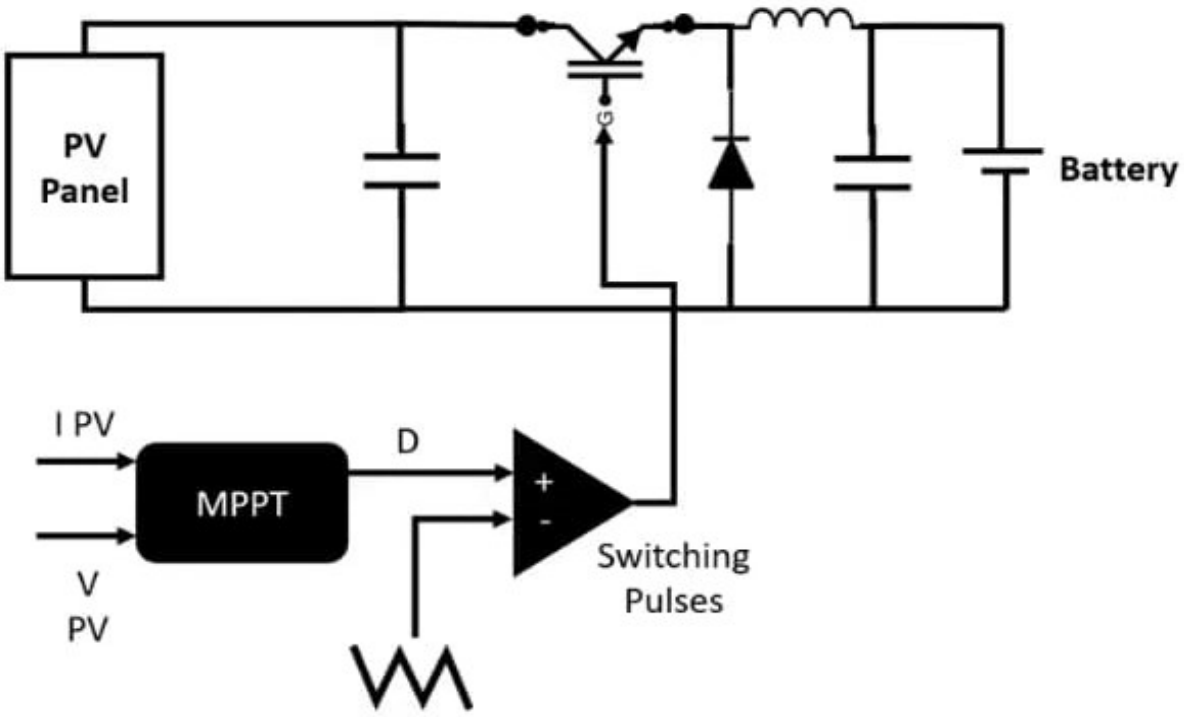
SOC


Voltage M1
405
400
395
390
385
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Offset=0

RESULT:

Thus, the bidirectional energy flow between the EV and the grid in V2G and G2V modes was efficiently demonstrated using MATLAB - Simulink. The Grid stability was improved during peak demand by V2G mode, while the battery effectively charged in G2V mode.

CIRCUIT DIAGRAM:



EX. NO. DATE:	DESIGN AND DEMONSTRATE SOLAR PV BASED CHARGING STATION	

AIM:

To design and simulate a Solar Photovoltaic (PV)-based charging station using MATLAB/Simulink

SOFTWARE REQUIRED:

MATLAB software with Simulink toolbox

THEORY

The **Solar PV-based EV Charging Station** harnesses solar energy to efficiently charge an electric vehicle (EV) battery. The system incorporates several components, including a **Solar PV Array**, **Maximum Power Point Tracking (MPPT)**, a **DC-DC Boost Converter**, and an **EV Battery Charging System**.

- The **Solar PV Array** captures sunlight and converts it into direct current (DC) electricity. The amount of power generated depends on solar irradiance and temperature, which vary throughout the day. This provides the initial energy source for the system.

- To extract the maximum possible power from the PV array, the **MPPT Controller** optimizes the operation of the solar panels. Since the output power from a solar panel is nonlinear, the MPPT adjusts the duty cycle of the connected boost converter to track the **Maximum Power Point (MPP)**. The most common technique used is **Perturb and Observe (P&O)**, which continuously monitors power and voltage changes to determine the optimal operating point.

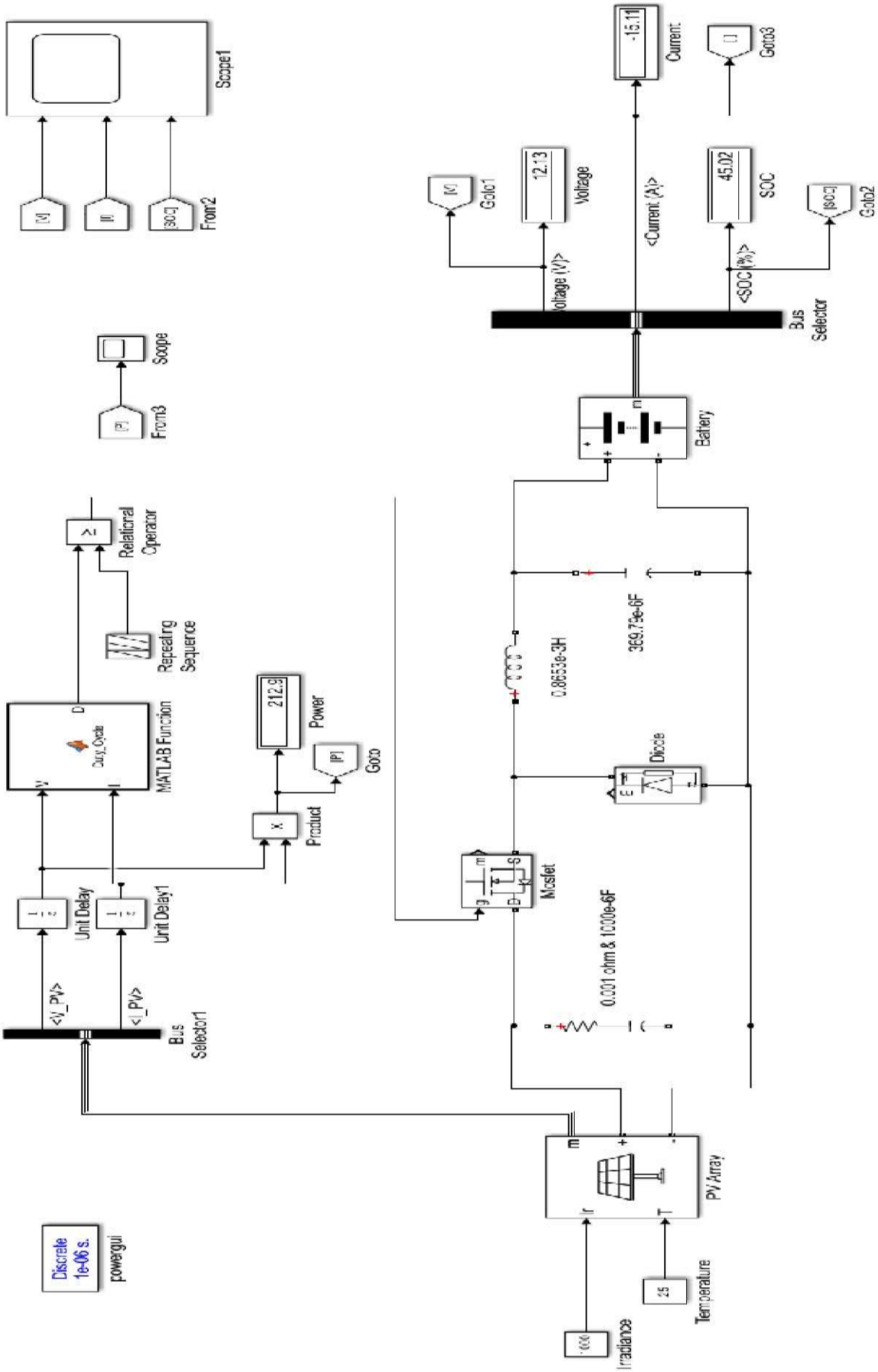
- The **DC-DC Boost Converter** steps up the voltage from the PV array to a level suitable for charging the EV battery. The MPPT controller modulates the duty cycle of the boost converter, ensuring the voltage is increased to the required level (e.g., 300V) for effective charging.

- The EV battery is charged using a **Constant Current / Constant Voltage (CC/CV)** charging method. Initially, the battery is charged with a constant current until it reaches a specified voltage (e.g., 300V). Once the voltage limit is reached, the charging switches to constant voltage mode, where the voltage is maintained, and the current gradually decreases as the battery nears full charge.

- The system continuously monitors the **State of Charge (SOC)** of the EV battery to ensure safe and efficient charging. The charging rate is adjusted based on the available solar power and the battery's current charge level. In cases of insufficient sunlight, the system may rely on stored energy or grid power (if integrated).

- The **Solar PV-based EV Charging Station** combines solar energy harvesting, MPPT optimization, voltage step-up via a boost converter, and a CC/CV charging algorithm to efficiently charge an EV battery. This setup provides a clean, renewable, and sustainable solution for EV charging, reducing dependence on traditional grid electricity and promoting the use of solar energy.

SIMULATION DIAGRAM:



SIMULATION PROCEDURE:

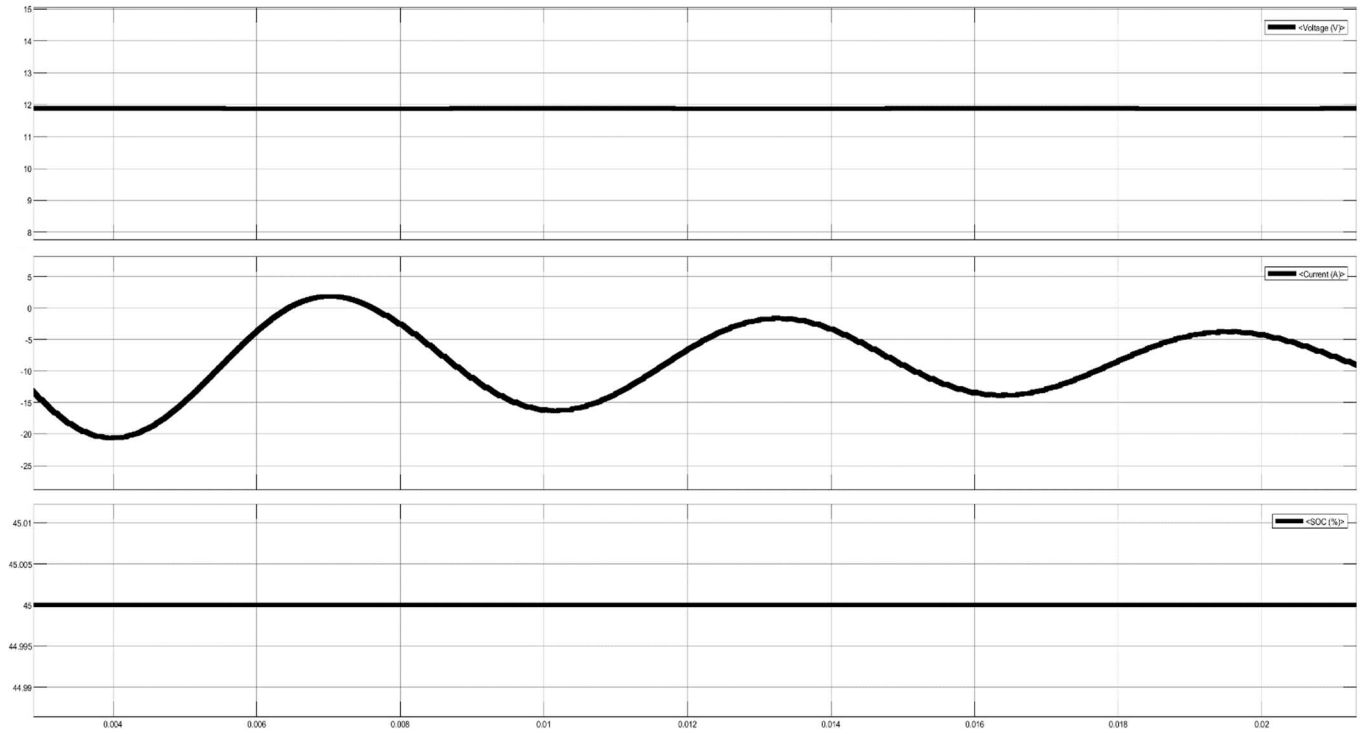
1. Open MATLAB and create a new Simulink model
2. Add a PV Array to convert solar energy into DC power
3. Implement MPPT with a MATLAB Function block using the Perturb and Observe algorithm
4. Connect a DC-DC Boost Converter to step up the voltage for battery charging
5. Integrate an EV Battery block and use Constant Current / Constant Voltage (CC/CV) charging
6. Monitor key parameters (PV power, battery voltage, charging current, SOC) using Scope blocks
7. Set the simulation solver and time step (e.g., ode4, 1e-6 s)
8. Run the simulation to evaluate system performance and analyze results

PROGRAM:

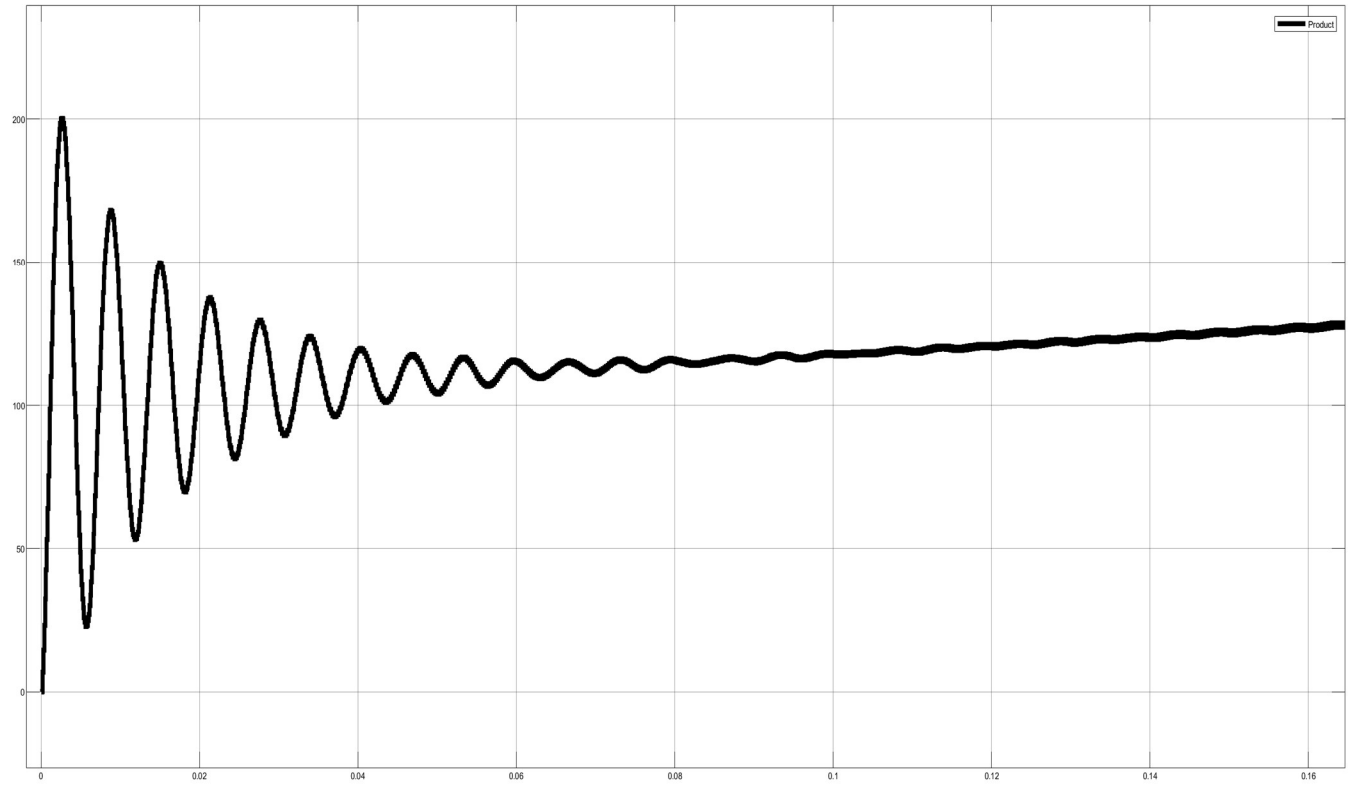
```
function D=DutyRatio(V,I)
Dmax=0.95;
Dmin=0;
Dinit=0.95;
deltaD=0.0001;
persistent Vold Pold Dold;
dataType='double';
if isempty(Vold)
   Vold=0;
   Pold=0;
   Dold=0;
end
P=V*I;
dV=V-Vold;
dP=P-Pold;
if dP ~=0
   if dP<0
      if dV<0
         D= Dold - deltaD;
      else
         D= Dold+deltaD;
      end
   else
      if dV<0
         D= Dold+deltaD;
      else
         D=Dold-deltaD;
      end
   end
else D=Dold;
end
if D>=Dmax || D<=Dmin
   D=Dold;
end
Dold=D;
Vold=V;
Pold=P;
```

WAVEFORMS:

Scope:



Power:



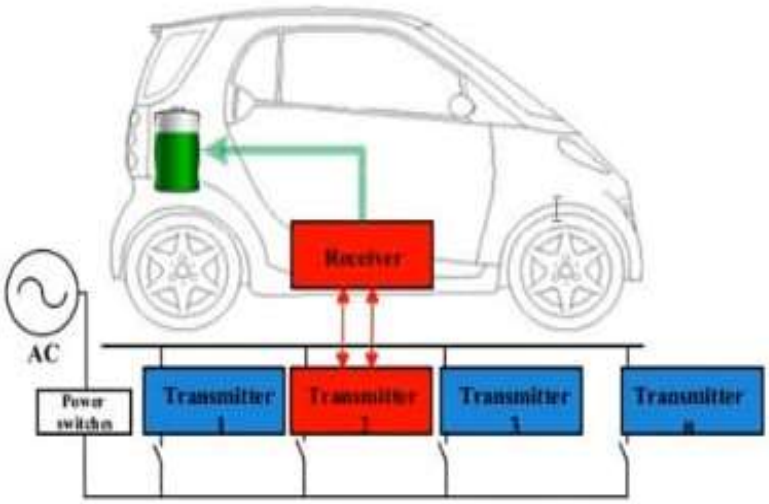
CALCULATIONS:

RESULT:

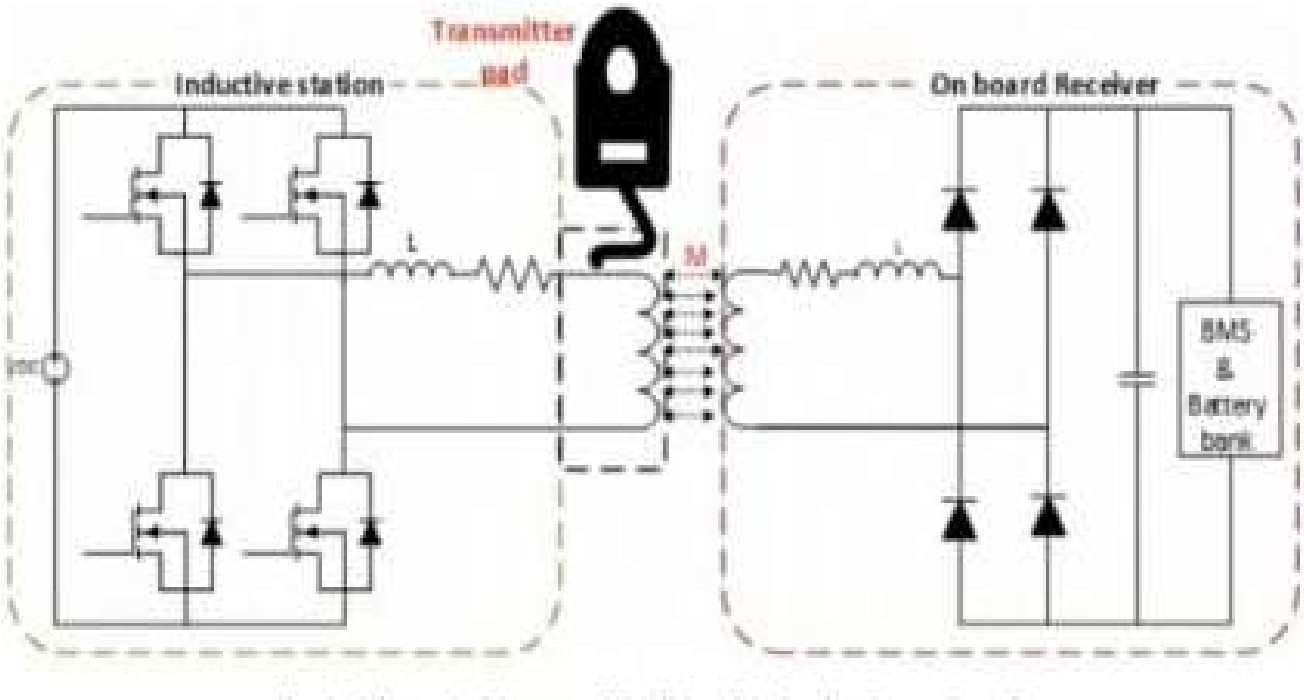
Thus the **Solar PV-based EV Charging Station** was successfully designed and demonstrated using MATLAB/Simulink, where solar energy was converted into DC power and optimized using the **MPPT** algorithm.

CIRCUIT DIAGRAM:

DWPT system with parallel transmitters:



Traditional Inductive Power Transfer:



EX. NO. DATE:	SIMULATE AND INFER WIRELESS POWER CHARGING STATION FOR EV CHARGING	

AIM:

To simulate the working of a Wireless Power Transfer (WPT) system for Electric Vehicle (EV) charging using MATLAB/Simulink

SOFTWARE REQUIRED:

MATLAB software with Simulink toolbox

THEORY:

Wireless Power Transfer (WPT) using **inductive coupling** is a method of transferring energy from a power source to a device without physical contact, relying on electromagnetic induction. This principle is widely used in applications like **EV charging stations** for electric vehicles (EVs).

1. Transmitter Coil (Primary Coil):

- The transmitter coil is connected to an AC power source. When an alternating current flows through the coil, it generates a time-varying magnetic field around it.
- This magnetic field extends into the surrounding space, creating a region where energy can be transferred wirelessly.

2. Receiver Coil (Secondary Coil):

- The receiver coil in the EV is placed within this magnetic field. As the magnetic field from the transmitter coil changes, it induces an electromotive force (EMF) in the receiver coil, according to Faraday's Law of Induction.
- This induced EMF generates an electric current in the receiver coil.

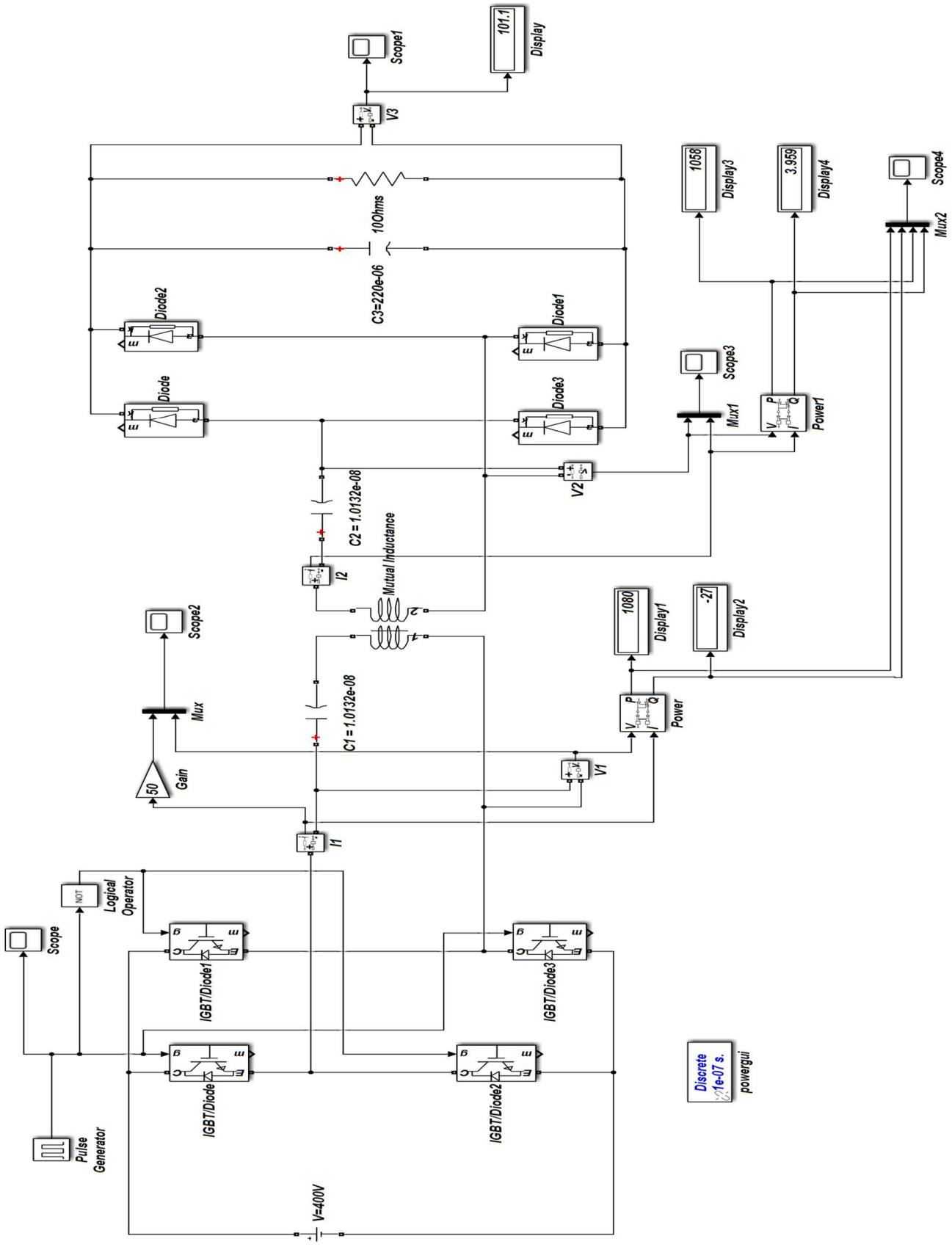
3. Power Conversion and Delivery:

- The induced current in the receiver coil is rectified (converted from AC to DC) using a rectifier circuit.
- The DC power is then used to charge the EV battery.

4. Efficiency Factors:

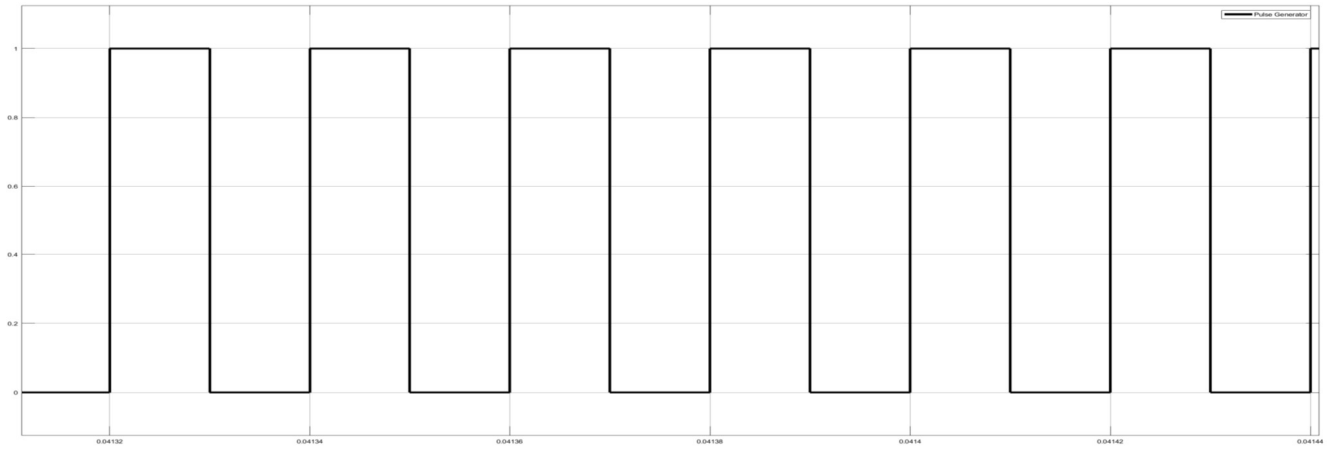
- The efficiency of power transfer in WPT depends on the distance between the coils, alignment, and the strength of magnetic coupling. The closer and more precisely aligned the coils are, the stronger the coupling, and the higher the energy transfer efficiency.
- The system can be optimized by tuning the coils to the same resonant frequency, which maximizes power transfer efficiency.

SIMULATION DIAGRAM:

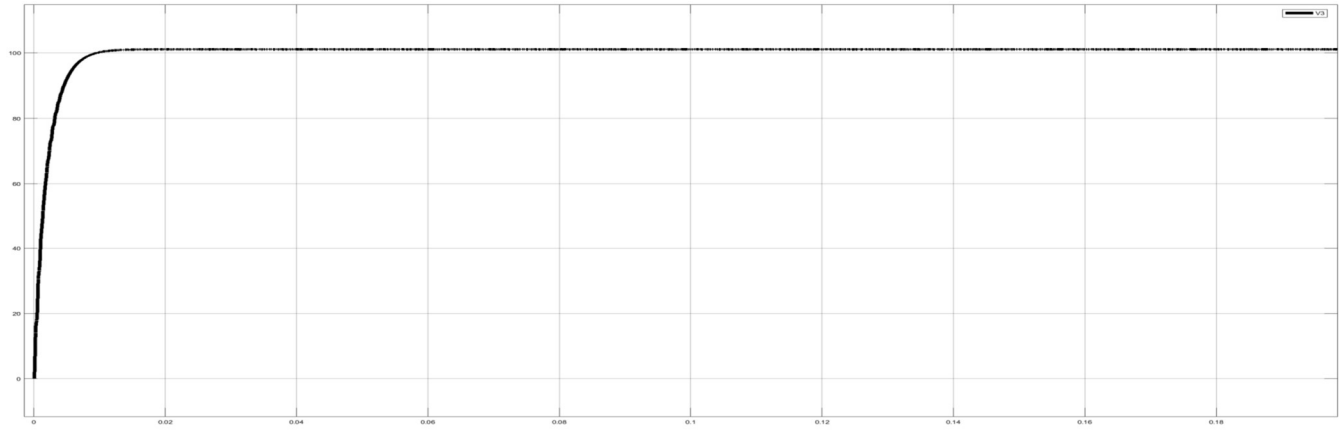


WAVEFORMS:

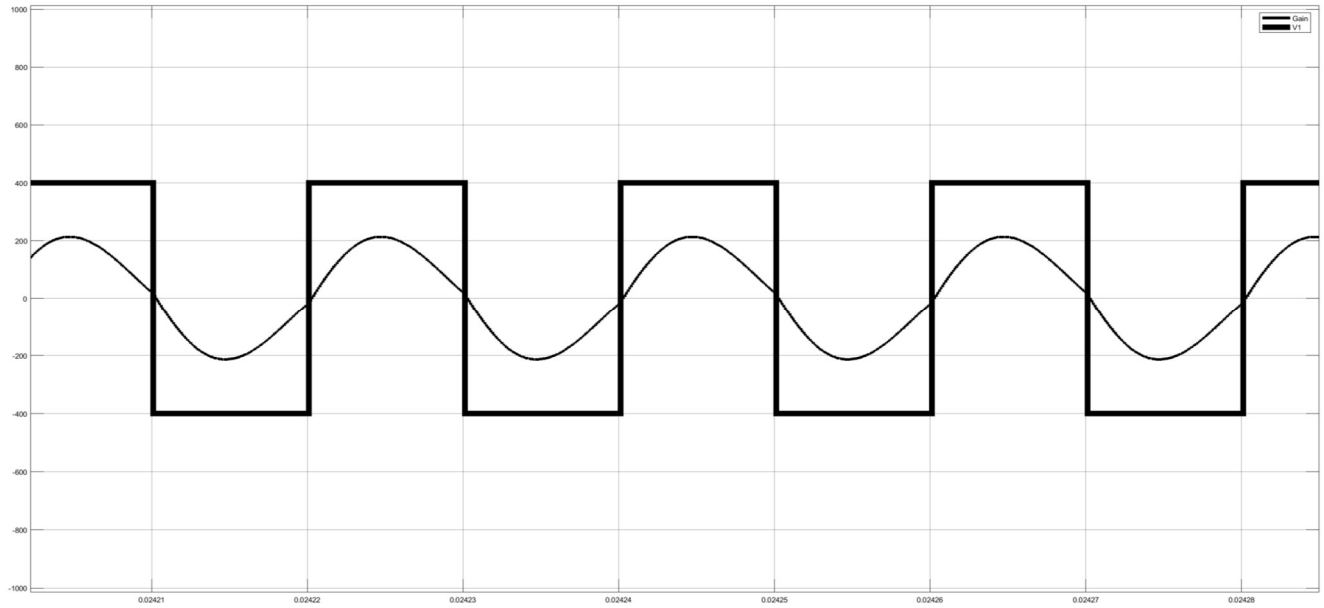
Pulse Generator



Scope 1

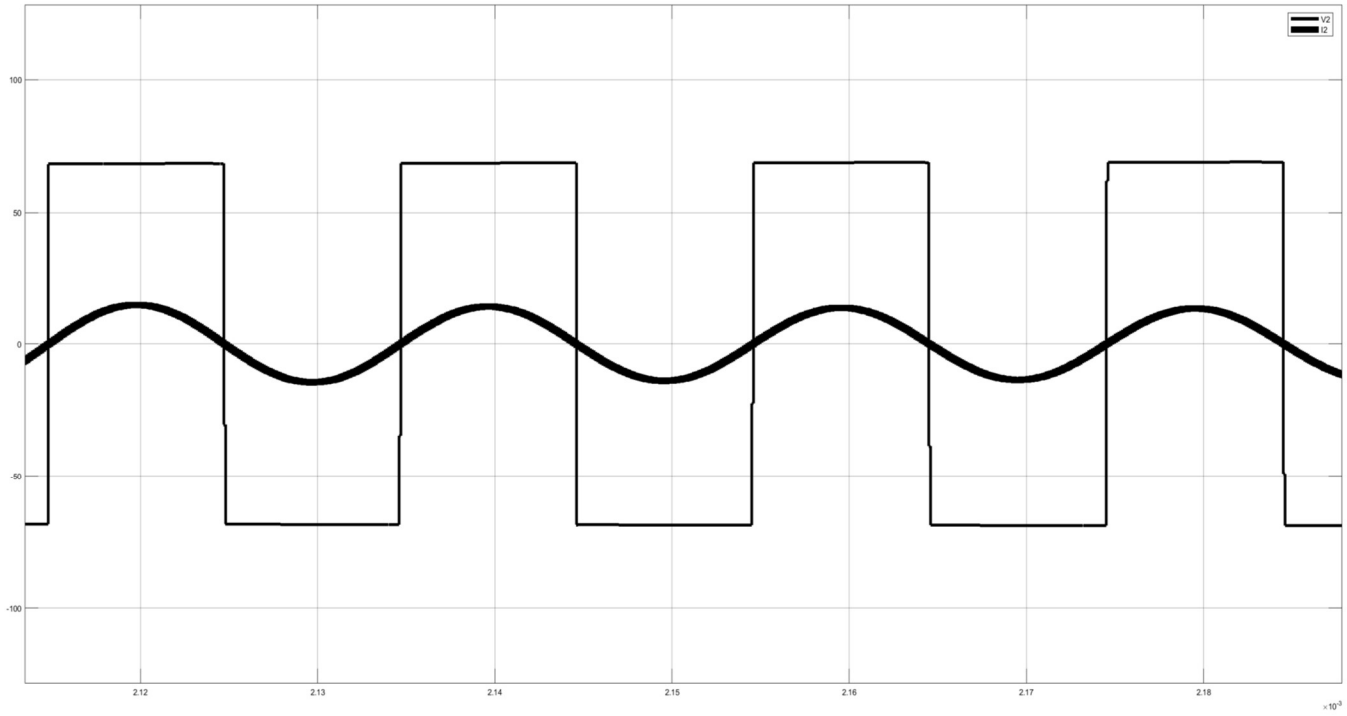


Scope 2

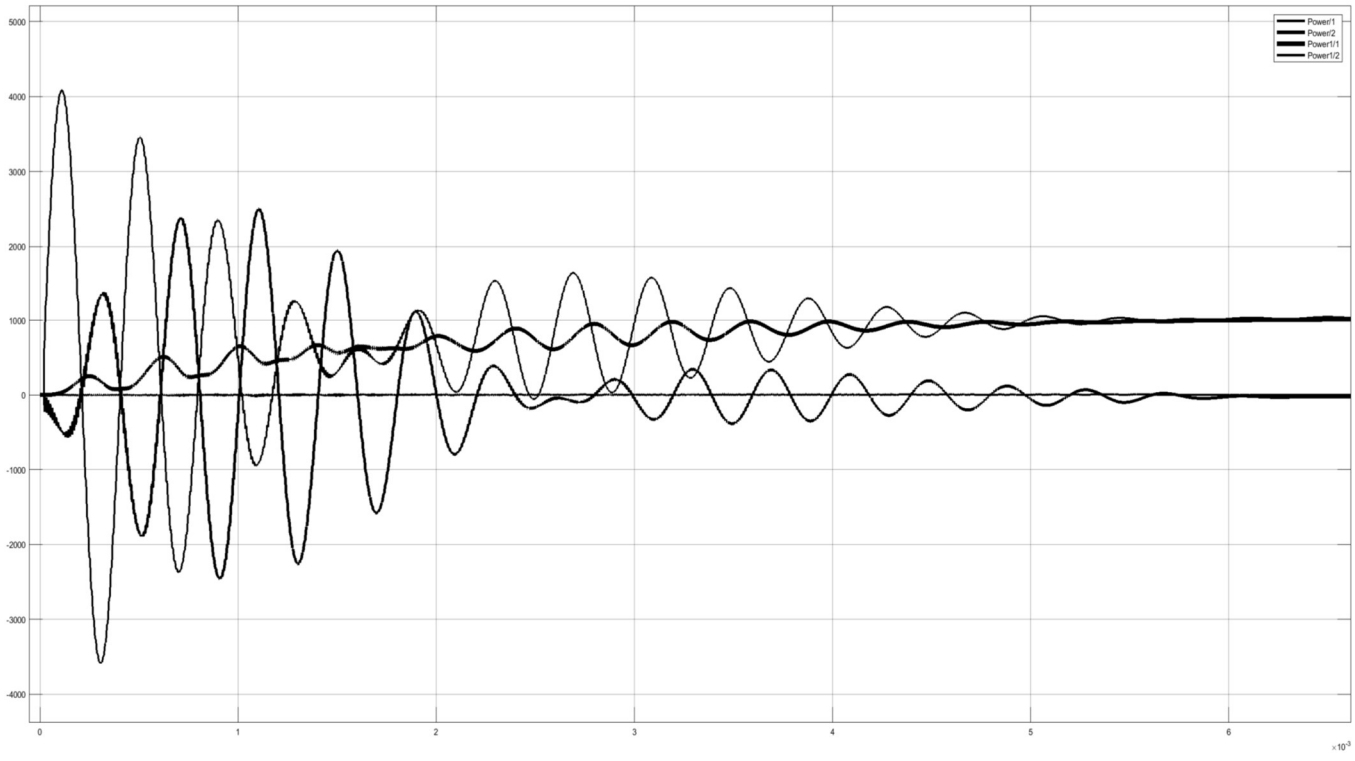


WAVEFORMS:

Scope 3



Scope 4



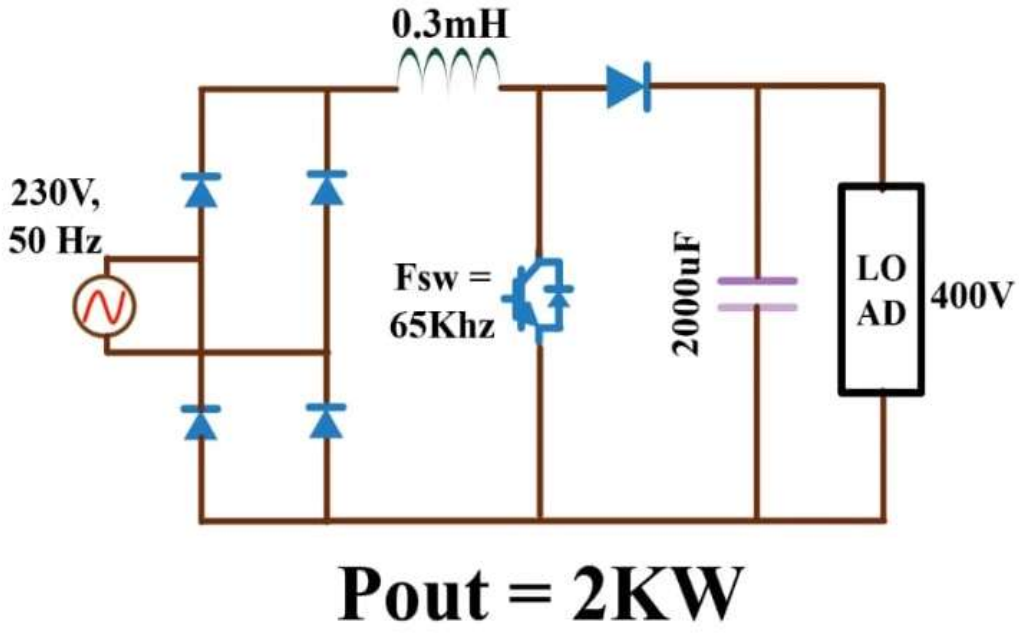
SIMULATION PROCEDURE:

1. Open MATLAB and create a new Simulink model
2. Model transmitter and receiver coils with **AC voltage source**, **Inductor** blocks, and **Mutual Inductance**.
3. Use **full-wave rectifier** to convert AC to DC and connect to **Battery** block for charging.
4. Implement **DC-DC converter** and **feedback loop** with **PID controller** to regulate charging.
5. Add **Voltage/Current Measurement** blocks and set simulation parameters (frequency, inductance, distance, charging power).
6. Configure **simulation time** and select **solver (like ode45)**, then run simulation to evaluate power transfer efficiency and charging performance.

RESULT:

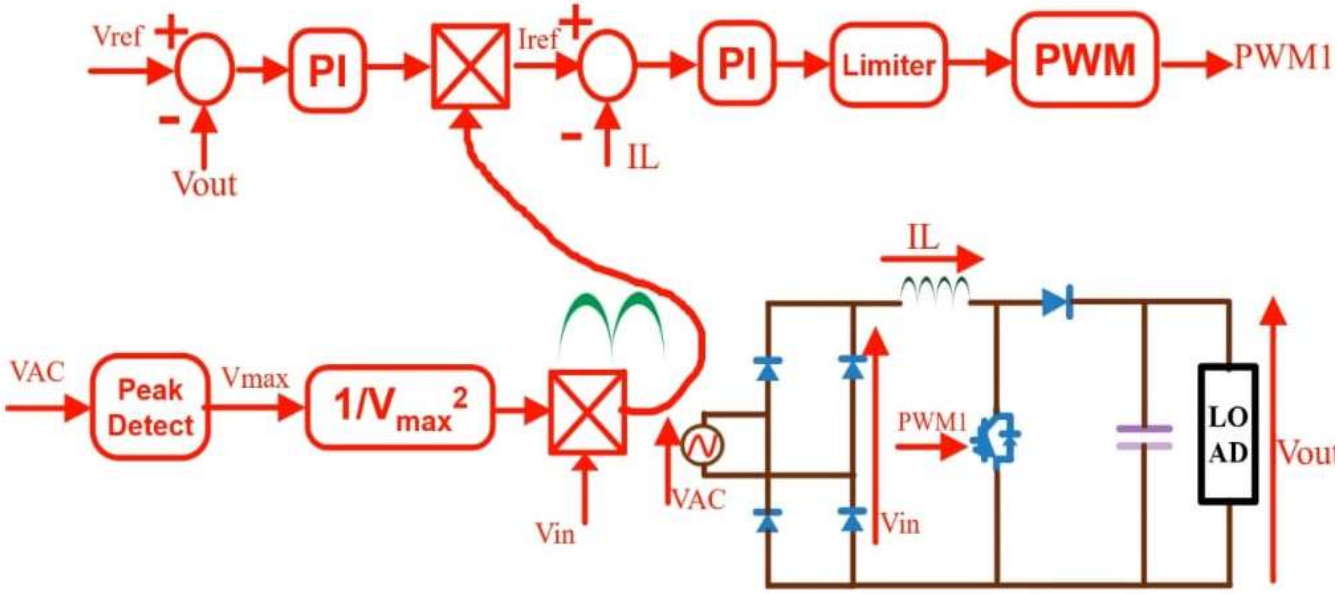
Thus, the wireless power charging station for EV charging has been simulated and analyzed using MATLAB - Simulink.

CIRCUIT DIAGRAM:



Pout = 2KW

CONTROLLER BLOCK DIAGRAM:



EX. NO. DATE:	SIMULATION OF BOOST CONVERTER BASED POWER FACTOR CORRECTION	

AIM:

To simulate and analyze the performance of a Boost Converter-based Power Factor Correction (PFC) circuit using MATLAB/Simulink

SOFTWARE REQUIRED:

MATLAB software with Simulink toolbox

THEORY:

- The Boost Converter-based PFC circuit is designed to improve the power factor by aligning the input current with the input voltage waveform, thus reducing the phase difference.
- AC input voltage is first converted to a pulsating DC using a rectifier, which then serves as the input to the Boost Converter circuit.
- The Boost Converter consists of an inductor, a diode, a capacitor, and a MOSFET switch. During operation, the MOSFET alternates between on and off states to control the inductor's energy storage and release:
  - When the MOSFET is on, the inductor stores energy from the input source.
  - When the MOSFET turns off, the inductor releases this stored energy to the output through the diode, resulting in a boosted output voltage.

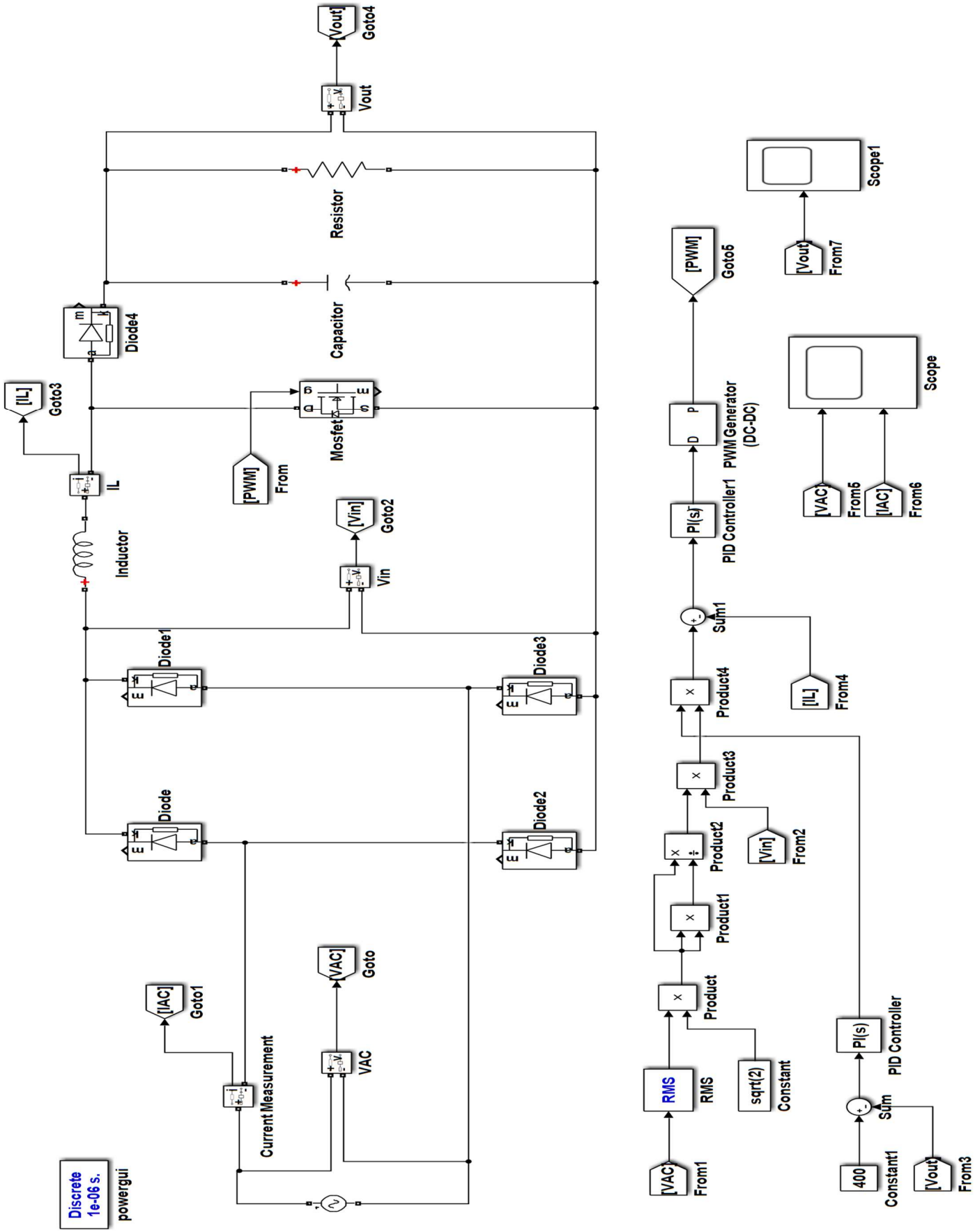
- **Power Factor (PF)** is a measure of how effectively electrical power is being used. It is defined as the ratio of real power (useful power) to apparent power (total power), or mathematically:

$$\textbf{Power Factor} = \cos(\phi)$$

where ϕ is the phase angle difference between the input voltage and current.

- A **high power factor** (close to 1) means that most of the power is being used effectively, with minimal losses, while a low power factor indicates wasted power due to reactive components (inductive or capacitive loads).
- Thus, the core of the PFC operation lies in shaping the input current to follow the input voltage waveform. By dynamically adjusting the duty cycle of the MOSFET, the input current can be controlled to create a sinusoidal waveform that is in phase with the rectified input voltage.
- A feedback control loop is employed to adjust the duty cycle based on real-time measurements of the input current and voltage, maintaining synchronization between the current and voltage waveforms.
- This adjustment helps achieve a high power factor (close to unity), reduces reactive power, and minimizes harmonic distortion in the input current.
- Overall, the Boost Converter-based PFC circuit ensures efficient power use, stable output voltage, and compliance with power quality standards by reducing losses and improving the power factor.

SIMULATION DIAGRAM:

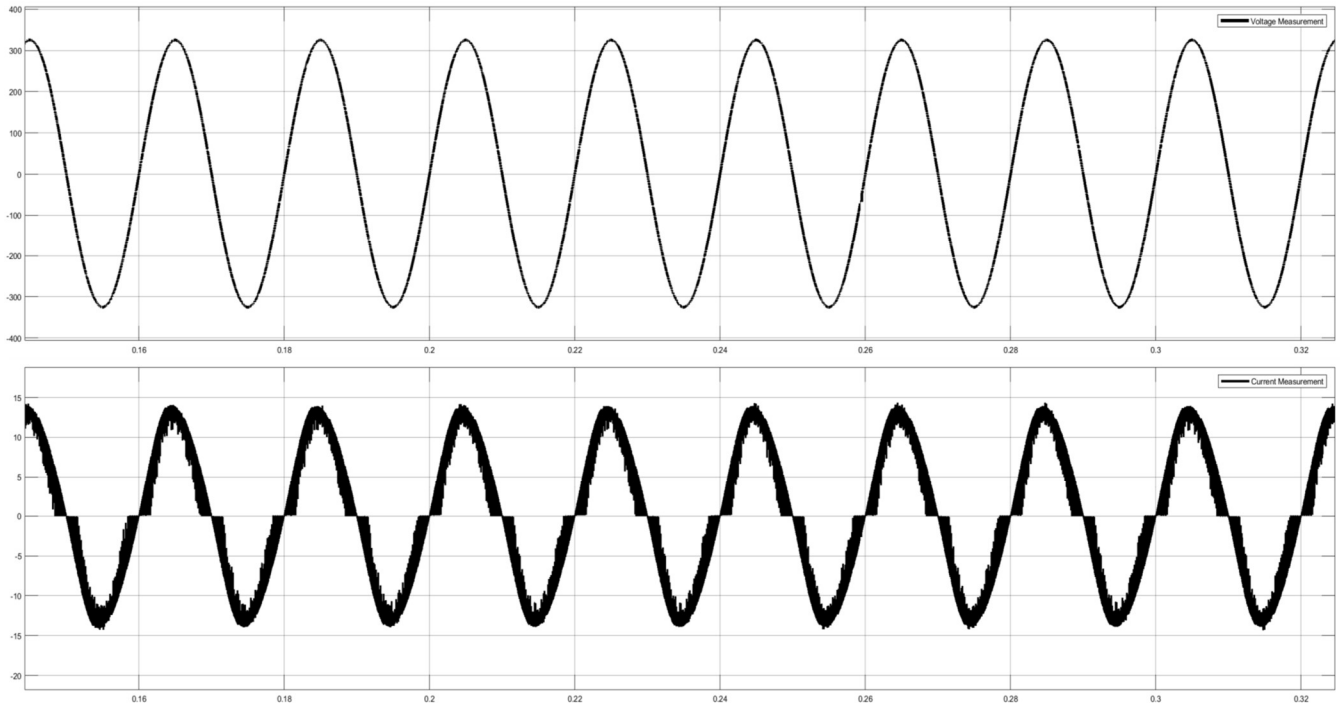


SIMULATION PROCEDURE:

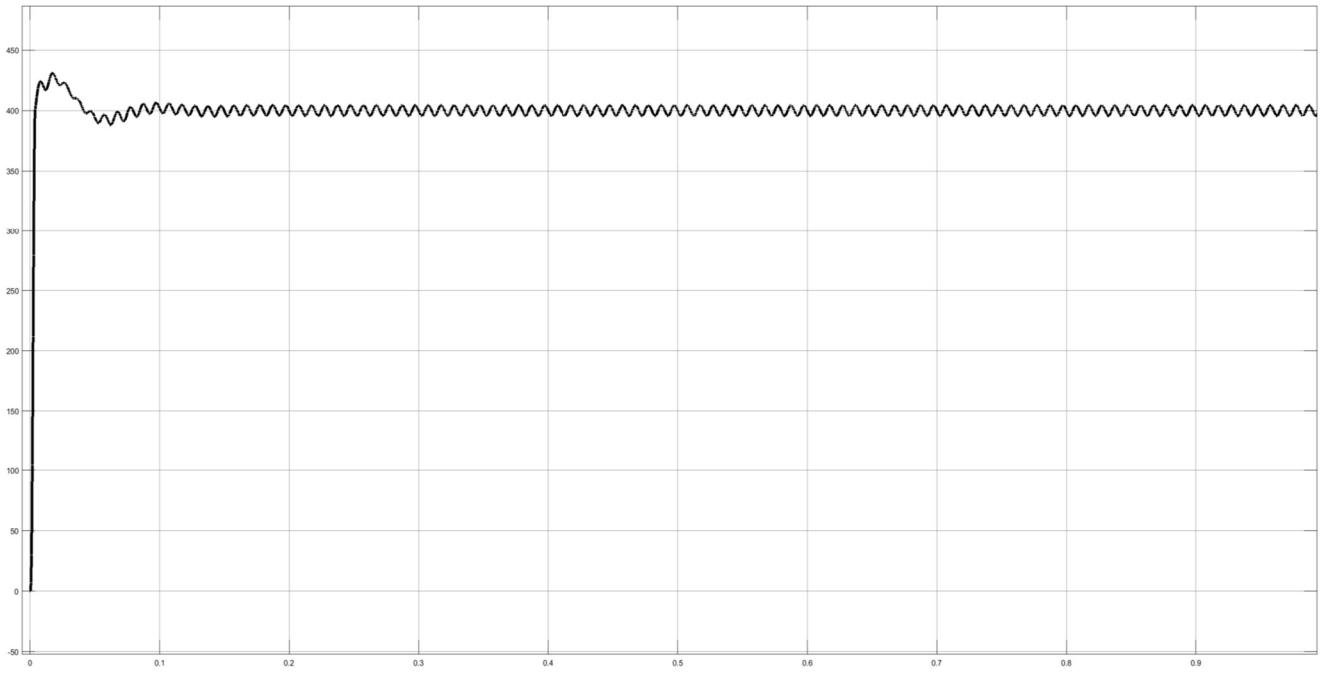
1. Open MATLAB/Simulink, create a new model, and add an AC voltage source with a rectifier to convert AC input to DC.
2. Connect an LC filter (optional) to reduce noise, then set up the Boost Converter with an inductor, diode, capacitor, and MOSFET.
3. Add a resistive load at the output to simulate the DC load.
4. Configure a PWM generator for the MOSFET, setting its frequency and enabling dynamic duty cycle control.
5. Set up a feedback loop with a current sensor, voltage sensor, and PI controller to adjust the duty cycle and achieve a sinusoidal input current in phase with the voltage.
6. Add an additional PI controller for output voltage regulation to maintain stable output despite input or load changes.
7. Run the simulation, observe the input/output waveforms, and verify that the input current is sinusoidal and in phase with the voltage, indicating effective power factor correction.

WAVEFORMS:

Input Voltage and Current:



Output Voltage:



RESULT:

Thus, the performance of a Boost Converter-based Power Factor Correction (PFC) circuit had been simulated and analyzed using MATLAB - Simulink.